

THE IMPACTS OF RAINFALL ON EMERGENCY RESPONSE TIMES

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HOUSTON FIRE DEPARTMENT

FRIDAY, APRIL 17, 2026

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OUTLINE

- I. FORMING PARTNERSHIPS
- II. DATA SOURCES
- III. RESEARCH AND ANALYSIS
- IV. POTENTIAL

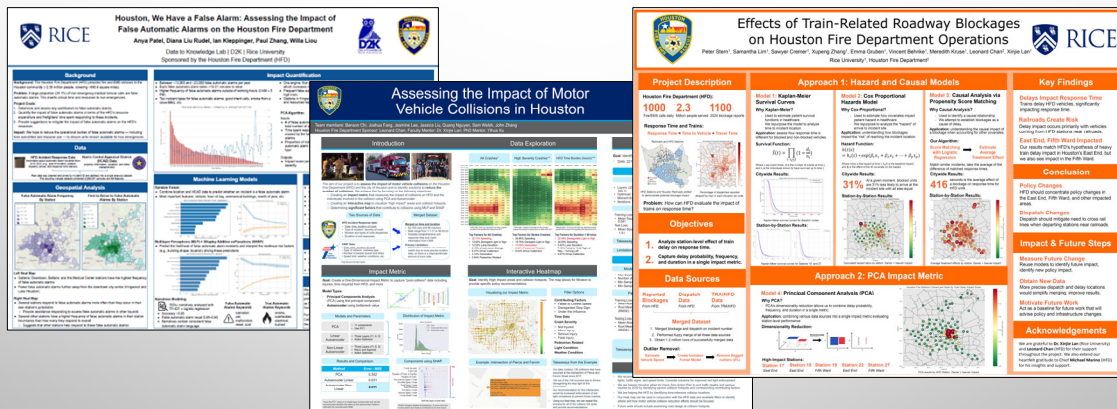
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FORMING PARTNERSHIPS



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PROJECTS WITH RICE DATA-TO-KNOWLEDGE (D2K) PROGRAM



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PROJECTS WITH RICE DATA-TO-KNOWLEDGE (D2K) PROGRAM

- SPRING 2009: OPTIMIZING PLACEMENT OF EMERGENCY VEHICLES FOR THE HOUSTON FIRE DEPARTMENT
- FALL 2009: A DATA-DRIVEN APPROACH FOR MITIGATING INEFFICIENT DISPATCHING IN THE HOUSTON FIRE DEPARTMENT
- SPRING 2020: EMERGENCY RESPONSE DEMOGRAPHIC-BASED RISK ASSESSMENT FOR HOUSTON FIRE DEPARTMENT
- FALL 2020: THE IMPACT OF COVID-19 ON THE HOUSTON FIRE DEPARTMENT
- SPRING 2021: OPTIMIZING HOUSTON'S EMERGENCY RESPONSE FLEET PROFILE
- FALL 2021: ANALYZING FIRE INSPECTIONS FOR THE HOUSTON FIRE DEPARTMENT
- SPRING 2022: MODELING FIRE RISK IN THE CITY OF HOUSTON
- FALL 2022: UNDERSTANDING CONTRIBUTING FACTORS TO MOTOR VEHICLE INCIDENTS IN THE HOUSTON FIRE DEPARTMENT

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PROJECTS WITH RICE DATA-TO-KNOWLEDGE (D2K) PROGRAM

- SPRING 2023: IDENTIFYING MOTOR VEHICLE COLLISION PREVENTION OPPORTUNITIES IN THE HOUSTON FIRE DEPARTMENT
- FALL 2023: ASSESSING THE IMPACT OF MOTOR VEHICLE COLLISIONS IN THE CITY OF HOUSTON
- **FALL 2024: EFFECTS OF RAINFALL ON EMERGENCY RESPONSE TIMES FOR THE HOUSTON FIRE DEPARTMENT; ENVIRONMENTAL AND DEMOGRAPHIC FACTORS IMPACTING HOUSTON FIRE DEPARTMENT EMERGENCY RESPONSE DEMANDS**
- SPRING 2025: EVALUATION OF TRAIN DELAY MITIGATION STRATEGIES USING ALTERNATIVE DISPATCH PROTOCOLS; DETERMINING THE IMPACT OF RAIL TRAFFIC ON FIRE AND EMS RESPONSE TIMES
- FALL 2026: HOUSTON, WE HAVE A FALSE ALARM: ASSESSING THE IMPACT OF FALSE AUTOMATIC ALARMS ON THE HOUSTON FIRE DEPARTMENT

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HARRIS COUNTY FLOOD CONTROL DISTRICT

THE FLOOD CONTROL DISTRICT IS A SPECIAL PURPOSE DISTRICT CREATED BY THE TEXAS LEGISLATURE IN 1937 AND GOVERNED BY HARRIS COUNTY COMMISSIONERS COURT. IT WAS CREATED IN RESPONSE TO DEVASTATING FLOODS THAT STRUCK THE REGION IN 1929 AND 1935. THE FLOOD CONTROL DISTRICT'S JURISDICTIONAL BOUNDARIES ARE SET TO COINCIDE WITH HARRIS COUNTY, A COMMUNITY OF MORE THAN 4.7 MILLION PEOPLE (2020 CENSUS) THAT INCLUDES THE CITY OF HOUSTON. THE OTHER BOUNDARIES IN WHICH WE OPERATE – THOSE PROVIDED BY NATURE – ARE THE 23 PRIMARY WATERSHEDS WITHIN HARRIS COUNTY'S 1,777 SQUARE MILES. EACH WATERSHED HAS ITS OWN INDEPENDENT FLOODING PROBLEMS. EACH WATERSHED PRESENTS UNIQUE CHALLENGES.

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FLOOD WARNING SYSTEM



THE [HARRIS COUNTY FLOOD CONTROL DISTRICT'S FLOOD WARNING SYSTEM](#) MEASURES RAINFALL AMOUNTS AND MONITORS WATER LEVELS IN BAYOUS AND MAJOR STREAMS ON A REAL-TIME BASIS TO INFORM YOU OF DANGEROUS WEATHER CONDITIONS. THE SYSTEM RELIES ON 200 [GAGE STATIONS](#) STRATEGICALLY PLACED THROUGHOUT HARRIS COUNTY BAYOUS AND THEIR [TRIBUTARIES](#). THE STATIONS CONTAIN [SENSORS](#) THAT TRANSMIT VALUABLE DATA DURING TIMES OF HEAVY RAINFALL AND DURING TROPICAL STORMS AND HURRICANES. SOME GAGES ALSO MEASURE WIND SPEED AND DIRECTION, BAROMETRIC PRESSURE, AIR TEMPERATURE, ROAD TEMPERATURE AND HUMIDITY.

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HISTORY OF FLOOD WARNING SYSTEM

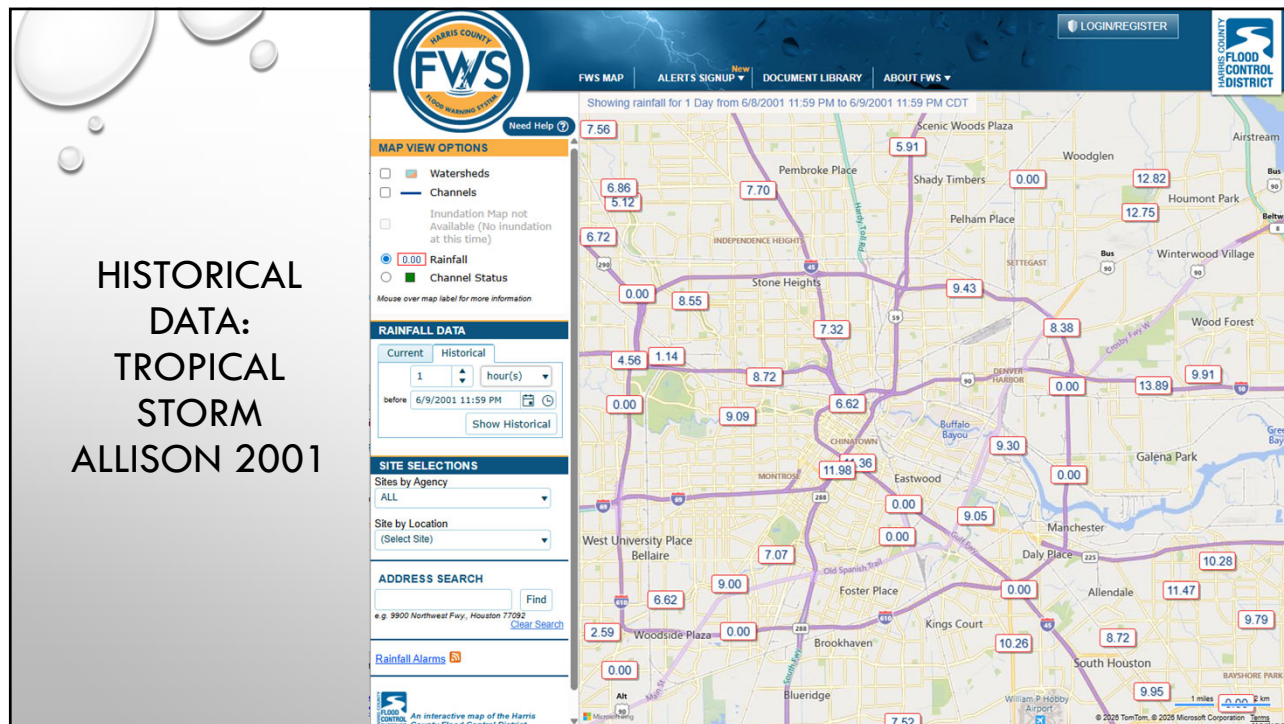
THE INSTALLATION OF THE GAGE SYSTEM BEGAN IN 1982 UNDER THE DIRECTION OF THE FLOOD CONTROL DISTRICT AND INITIALLY INCLUDED 13 GAGE STATIONS. THE FIRST TEST OF THIS SYSTEM OCCURRED DURING HURRICANE ALICIA IN 1983, AND THE SYSTEM PROVIDED INVALUABLE INFORMATION, INCLUDING RAINFALL TOTALS AND BAYOU/STREAM LEVEL DATA THAT WAS NOT PREVIOUSLY AVAILABLE. FROM 1983 TO 2021, THE NUMBER OF GAGE STATIONS INCREASED TO TODAY'S TOTAL OF 188, EXPANDING THE ABILITY TO MONITOR AND GATHER INFORMATION FROM MANY MORE BAYOUS AND STREAMS.

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HOW THE FLOOD WARNING SYSTEM WORKS

- WHEN IT BEGINS TO RAIN, DATA-COLLECTING SENSORS AT EACH GAGE STATION TRANSMIT RAINFALL AMOUNTS VIA RADIO FREQUENCY EVERY TIME 0.04 INCHES OF RAIN IS MEASURED BY THE SENSOR. SENSORS THAT TRANSMIT BAYOU/STREAM LEVELS REPORT EVERY 0.10 FOOT RISE IN WATER LEVELS.
- THE SENSORS TRANSMIT INFORMATION TO FOUR PRIMARY REPEATERS LOCATED IN THE HUFFMAN, CLODINE, LEAGUE CITY, AND TOMBALL AREAS. THE REPEATERS THEN RELAY THE DATA TO PRIMARY AND BACK-UP BASE STATIONS LOCATED AT HOUSTON TRANSTAR AND AT THE HARRIS COUNTY HOUSEHOLD AND HAZARDOUS WASTE FACILITY. THE INFORMATION IS MONITORED DAILY BY SOFTWARE AND BY FLOOD CONTROL DISTRICT STAFF TO ENSURE THE GAGES ARE PROPERLY FUNCTIONING AND TRANSMITTING ACCURATE DATA.

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PUBLIC INFORMATION REQUEST

The following Public data was submitted on Tuesday, July 30, 2024 4:38:06 PM. It was submitted via the Flood Warning System Public web contact form.

First Name:	Leonard
Last Name:	Chan
Email:	leonard.chan@houston.tx.gov
Phone:	7138580819
Address:	500 Jefferson Houston, 500 Jefferson - 77002
Sending to Agency:	Harris County Flood Control District
Sub-Division Type:	
I am a:	Govt
Preferred Contact Method:	ByEmail

Subject: Data Request

I am Leonard Chan, the accreditation manager with Houston Fire Department, and we are looking to conduct a study on our historic response times, and how they may be affected by rainfall. This would be in collaboration with researchers at Rice University. The data we are looking for is high resolution data (in 15 minute intervals) on rainfall for gauges across the city. Would there be an API available for these data?

Leonard,

We have some historical rainfall datasets available to download. How far back would you like the data? I can send you a link to download.

Jeremy Justice
Hydrologic Operations Manager
Harris County Flood Control District
9900 Northwest Freeway | Houston, Texas 77092

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DATA SOURCES

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HARRIS COUNTY FLOOD WARNING SYSTEM SENSORS

- SENSORS: 190
- START TIME: 2022-01-01 00:00:00
- END TIME: 2024-07-01 00:00:00
- INTERVAL: 5 MINUTES

	A	B	C	D	E	F	G	H	I
1	HCFCD								
2	Start time: 2022-01-01 00:00:00								
3	End time: 2024-07-01 00:00:00								
4	All timestamps are in local time.								
5	Interval: 5 min								
6	Null interval fill method: fill with zero value								
7									
8		Site	43020	43021	43022	43023	43024	43025	43026
9		Sensor	100	105	110	115	120	125	130
10		Site Name	Clear Lake 2	Marys Cree	Clear Creek	Cowart Cree	Clear Creek	Chigger Cree	Clear Creek
11	Begin time	End time							
12	1/1/2022 0:00	1/1/2022 0:05	0	0	0	0	0	0	0
13	1/1/2022 0:05	1/1/2022 0:10	0	0	0	0	0	0	0
14	1/1/2022 0:10	1/1/2022 0:15	0	0	0	0	0	0	0
15	1/1/2022 0:15	1/1/2022 0:20	0	0	0	0	0	0	0
16	1/1/2022 0:20	1/1/2022 0:25	0	0	0	0	0	0	0
17	1/1/2022 0:25	1/1/2022 0:30	0	0	0	0	0	0	0
18	1/1/2022 0:30	1/1/2022 0:35	0	0	0	0	0	0	0
19	1/1/2022 0:35	1/1/2022 0:40	0	0	0	0	0	0	0
20	1/1/2022 0:40	1/1/2022 0:45	0	0	0	0	0	0	0
21	1/1/2022 0:45	1/1/2022 0:50	0	0	0	0	0	0	0
22	1/1/2022 0:50	1/1/2022 0:55	0	0	0	0	0	0	0
23	1/1/2022 0:55	1/1/2022 1:00	0	0	0	0	0	0	0
24	1/1/2022 1:00	1/1/2022 1:05	0	0	0	0	0	0	0
25	1/1/2022 1:05	1/1/2022 1:10	0	0	0	0	0	0	0
26	1/1/2022 1:10	1/1/2022 1:15	0	0	0	0	0	0	0
27	1/1/2022 1:15	1/1/2022 1:20	0	0	0	0	0	0	0
28	1/1/2022 1:20	1/1/2022 1:25	0	0	0	0	0	0	0
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30									

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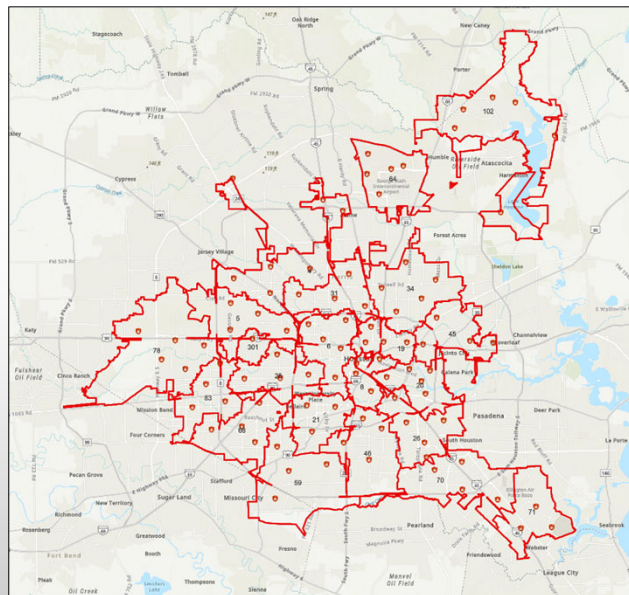
HOUSTON FIRE DEPARTMENT RESPONSE DATA

- INCIDENT DATA
 - 986,000+ INCIDENTS FROM JANUARY 2022 TO JUNE 2024
 - DATE, TIME, LOCATION
 - INCIDENT TYPE
- UNIT RESPONSE DATA
 - 1.9+ MILLION UNIT RESPONSES FROM JANUARY 2022 TO JUNE 2024
 - TURNOUT TIMES, ARRIVAL TIMES, CLEAR TIMES

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BACKGROUND DATA

- LOCATION OF RAIN GAUGES
- LOCATION OF STATIONS



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RESEARCH AND ANALYSIS

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PROJECT OBJECTIVES

- **EXAMINE THE IMPACT OF RAINFALL ON TRAVEL TIMES:** DETERMINE IF AND HOW RAINFALL IMPACTS HFD TRAVEL TIMES THROUGH VISUALIZATIONS AND MODELING. THIS WILL ALSO INCLUDE UNDERSTANDING HOW RAINFALL IMPACTS TRAVEL TIMES DIFFERENTLY IN DIFFERENT PARTS OF HOUSTON.
- **EXAMINE THE IMPACT OF UNIT TYPES AND OTHER VARIABLES ON TRAVEL TIMES:** ASSESS HOW THESE VARIABLES AFFECT TRAVEL TIMES THROUGH VISUALIZATIONS AND MODELING. THE ANALYSIS WILL INCLUDE EXPLORING IF LARGER, LESS MANEUVERABLE VEHICLES OR HEAVY RAIN CAUSE GREATER DELAYS.
- **PROPOSE ACTIONABLE CHANGES TO REDUCE TRAVEL TIME DELAYS:** BASED ON THE ANALYSIS OF CALL VOLUME AND TRAVEL TIMES, DEVELOP RECOMMENDATIONS FOR OPERATIONAL IMPROVEMENTS. THIS MAY INCLUDE OPTIMIZING UNIT DEPLOYMENT, ENHANCING NAVIGATION SYSTEMS TO AVOID WEATHER-RELATED OBSTACLES, AND REFINING RESPONSE PROTOCOLS TO ACCOUNT FOR WEATHER CONDITIONS, ULTIMATELY REDUCING TRAVEL TIME DELAYS.

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LITERATURE REVIEW

- **“WEATHER AND TEMPORAL FACTORS ASSOCIATED WITH USE OF EMERGENCY MEDICAL SERVICES” BY RAMGOPAL ET AL. (2019)** – STUDY IN PENNSYLVANIA THAT ANALYZED THE IMPACTS OF WEATHER CONDITIONS (TEMPERATURE, WIND, PRECIPITATION) ON RESPONSE DELAYS. RESPONSE TIMES INCREASED SIGNIFICANTLY FOLLOWING RAINFALL AND SNOW DUE TO REDUCED VISIBILITY, SLICK ROAD CONDITIONS, AND SLOWER VEHICLE SPEEDS. INVESTIGATED 6-HOUR RAINFALL PERIODS, WHICH MAY NOT CAPTURE THE IMMEDIATE IMPACTS OF INTENSE, SHORT-TERM RAINFALL EVENTS.
- **“EFFECTS OF RAINFALL ON THE USE OF EMERGENCY MEDICAL SERVICES” BY SAN ET AL. (2020)** – DETAILED ANALYSIS IN ANKARA, TURKEY THAT STUDIES THE IMPACT OF SEVERITY OF RAINFALL (NON-RAINY, LIGHT RAIN, MODERATE RAIN, AND HEAVY RAIN) ON CALL DEMANDS. LIGHT AND MODERATE RAIN MAY INCREASE DEMAND BUT SEVERE RAINFALL MAY DECREASE RAINFALL AS PEOPLE TEND TO STAY INDOORS. APPLICABILITY MAY BE LIMITED BY REGIONAL CLIMATE DIFFERENCES. **AMERICAN HEART ASSOCIATION (2021)** FOUND THAT EXTREME WEATHER EVENTS INCLUDING HEAVY RAINFALL LEADS A TO A SURGE IN EMERGENCY CALLS.

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LITERATURE REVIEW

- **“AN INTEGRATED APPROACH TO MODELING THE IMPACT OF FLOODS ON EMERGENCY SERVICES: A CASE STUDY OF CALGARY, ALBERTA” BY TSANG AND SCOTT (2020)** – DEMONSTRATED INCREASED WATER LEVELS LED TO LONGER TRAVEL TIMES DUE TO HINDERED VEHICLE SPEEDS. FOCUS WAS ON FLOODING RATHER THAN RAINFALL. CITED A STUDY IN PICKENS COUNTY (AL) THAT USED GEOGRAPHICALLY WEIGHTED REGRESSION (GWR) AND GLOBAL REGRESSION ANALYSIS (GRA) TO FIND LIMITED INFRASTRUCTURE AND LONGER TRAVEL DISTANCES LED TO DELAYS IN RESPONSE DURING ADVERSE WEATHER. THIS IS WORSENERD BY UNITS WITH FEWER ROUTE OPTIONS.
- **“DETERMINING MINIMUM STAFFING LEVELS DURING SNOWSTORMS USING AN INTEGRATED SIMULATION, REGRESSION, AND RELIABILITY MODEL” BY KUNKEL AND MCCLAY (2012)** – USED OF SCENARIO-BASED MODELING TO ASSESS STAFFING NEEDS UNDER ADVERSE WEATHER CONDITIONS SUCH AS SNOW.

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LITERATURE REVIEW

- **“RESPONSE TIME EFFECTIVENESS: COMPARISON OF RESPONSE TIME AND SURVIVAL IN AN URBAN EMERGENCY MEDICAL SERVICES SYSTEM” BY BLACKWELL (2008)** – USES SURVIVAL ANALYSIS TO STUDY EMERGENCY RESPONSE TIME TO UNDERSTAND THE EFFECT OF RESPONSE TIME ON PATIENT SURVIVAL. HIGHER SURVIVAL RATES ARE EXPECTED FOR RESPONSE TIMES THAT IS LESS THAN 5 MINUTES. *DEMONSTRATES THE IMPORTANCE OF RESPONSE TIMES.*
- **“TRAFFIC INCIDENT DURATION ANALYSIS AND PREDICTION MODELS BASED ON THE SURVIVAL ANALYSIS APPROACH” BY LI (2015)** – DEVELOPED A MODEL TO PREDICT THE LENGTH OF THE TRAFFIC EFFECTS ON DIFFERENT TYPES OF INCIDENTS ON FREEWAYS. *CONCEPTS COULD BE USED TO PREDICT THE TRAFFIC EFFECT OF FLOODING.*

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LITERATURE REVIEW

- **“SPATIAL MODELLING OF EMERGENCY SERVICE RESPONSE TIMES” BY TAYLOR (2017)** – APPLIES ADVANCED METHODS FROM SPATIAL SURVIVAL ANALYSIS TO CONDUCT INSIGHTS INTO RESPONSE TIMES. THE MODEL ANALYZES HOW RESPONSE TIME VARIES WITH TIME OF DAY AND DIFFERENT PARTS OF A JURISDICTION. CONCLUDED THAT THERE ARE LOGISTICAL LIMITS IN HOW MUCH RESPONSE TIME CAN BE REDUCED.

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D2K STATION VISIT



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MERGING DATA

- INCIDENT-RELATED DATA CONNECTED THROUGH THE INCIDENT ID.
- RAINFALL DATA:
 - DATA AGGREGATED AT EACH SENSOR FOR EACH HOUR. AMOUNT OF RAINFALL DURING THE HOUR OF THE INCIDENT USED.
 - CLOSEST SENSOR SELECTED FOR THE LOCATION OF AN INCIDENT.

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DERIVED DATA

- ESTIMATED TRAVEL TIME CALCULATED USING THROUGH THE ONLINE TRAVELTIME API APPLICATION.
- CATEGORIES CREATED FOR DIFFERENT RAIN CATEGORIES ACCORDING TO AMERICAN METEOREOLOGICAL SOCIETY:
 - NO RAIN – 0.0 INCHES PER HOUR
 - LIGHT RAIN – 0.0 TO 0.1 INCHES PER HOUR
 - MODERATE RAIN – 0.1 TO 0.3 INCHES PER HOUR
 - HEAVY RAIN – >3.0 INCHES PER HOUR

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DERIVED DATA

- RUSH HOUR DESIGNATION: 7-10 AM AND 4-7 PM.
- INCIDENT TYPE CATEGORIZATIONS BASED ON CAD DESCRIPTIONS. 23 CATEGORIES CREATED.

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OUTLIER REMOVAL

- INCIDENTS MORE THAN 3 STANDARD DEVIATIONS AWAY FROM THE MEAN LATITUDE AND LONGITUDE.
- TRAVEL TIMES LISTED AS ZERO OR NEGATIVE.
- TRAVEL DURATION MORE THAN THREE STANDARD DEVIATIONS FROM THE MEAN.
- USE OF ISOLATION FOREST (ISOLATES UNUSUAL DATA POINTS BY CONSTRUCTING A TREE STRUCTURE WHERE ANOMALIES ARE CLOSER TO THE ROOT OF THE TREE) TO IDENTIFY OUTLIER TRAVEL TIMES RELATIVE TO ESTIMATED TRAVEL TIMES.

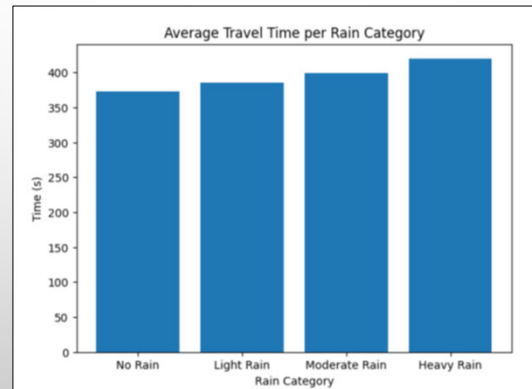
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DATA SET TO BE ANALYZED

- 619,000+ INCIDENTS WITH FIRST ARRIVING UNIT.
- 31 VARIABLES INCLUDING THE FOLLOWING:
 - INCIDENT LOCATION
 - INCIDENT TYPE
 - RESPONDING STATION
 - TRAVEL TIME
 - ESTIMATED DISTANCE TRAVELED
 - RAINFALL

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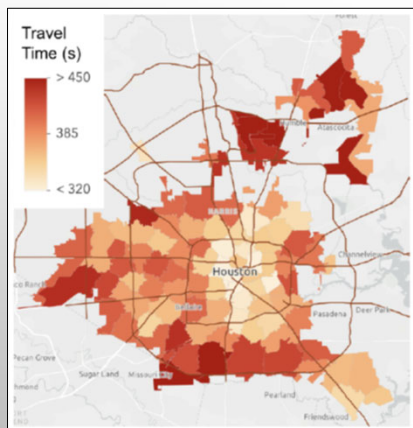
DATA EXPLORATION



HEAVIER RAIN CONDITIONS → HIGHER TRAVEL TIMES

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DATA EXPLORATION



- TRAVEL TIMES DIFFER BASED ON LOCATION.
- FASTER TRAVEL TIMES EXPECTED IN MORE DENSELY POPULATED AREAS.

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[illegible]

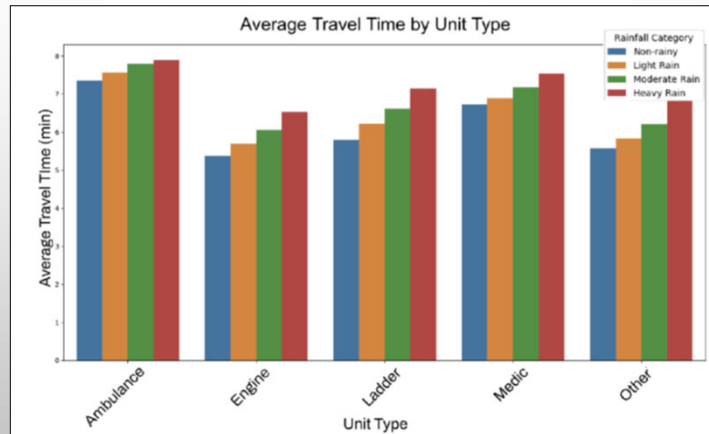
- 32

A bar chart titled "Average Estimated Distance by Unit Type". The y-axis is labeled "Distance (m)" and ranges from 0 to 3500 in increments of 500. The x-axis is labeled "Unit" and has five categories: Engine, Ambulance, Medic, Ladder, and Other. The bars are colored purple, red, green, orange, and blue respectively. The approximate values are: Engine (2600m), Ambulance (3700m), Medic (2950m), Ladder (2950m), and Other (3100m).

Unit	Average Estimated Distance (m)
Engine	2600
Ambulance	3700
Medic	2950
Ladder	2950
Other	3100

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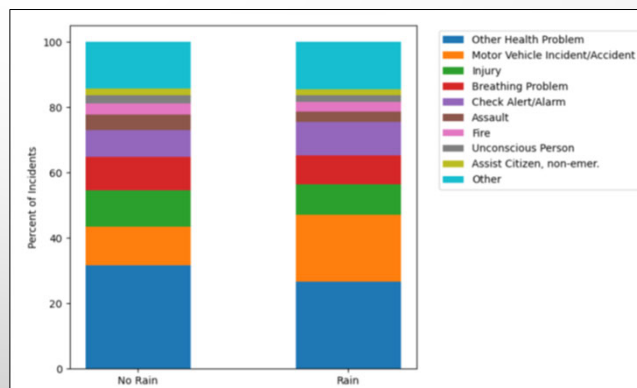
DATA EXPLORATION



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DATA EXPLORATION

- HIGHER PERCENTAGE OF MOTOR VEHICLE COLLISIONS IN RAIN CONDITIONS DUE TO RAIN CAUSING HAZARDOUS DRIVING CONDITIONS.



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DATA PROCESSING FOR MODELING

- **FEATURE SELECTION CONDUCTED** – REMOVES REDUNDANT/HIGHLY CORRELATED FEATURES.
 - CAD CODES USED INSTEAD OF NATIONAL FIRE INCIDENT REPORTING SYSTEM (NFIRS) CODES.
 - ESTIMATED TRAVEL DISTANCE VIA TRAVELTIME API USED INSTEAD OF ESTIMATED TRAVEL TIME.
- **ONE-HOT ENCODING** – CATEGORICAL FEATURES EXPANDED INTO SEPARATE BINARY FEATURES.
- **DATA SPLIT RANDOMLY INTO TRAINING (80%) AND TESTING (20%) SUBSETS.**

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MODELING

- **TIME-TO-EVENT ANALYSIS** – TO INFER VARIABLES' RELATIONSHIP WITH TRAVEL TIME.
- **REGRESSION MODELS** – TO PREDICT TRAVEL TIME AND INTERPRET IMPORTANCE OF FEATURES THAT CONTRIBUTE MOST TO TRAVEL TIME.

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TIME-TO-EVENT ANALYSIS

- USE OF **SURVIVAL ANALYSIS**, WHICH IS A STATISTICAL METHOD TO ANALYZE THE TIME IT TAKES FOR A SPECIFIC EVENT TO OCCUR. FOR THIS STUDY, THE EVENT CONSIDERED IS ARRIVAL ON SCENE.
- THE OUTPUT OF THIS ANALYSIS IS THE **HAZARD FUNCTION**, WHICH REPRESENTS THE INSTANTANEOUS RISK OF AN EVENT OCCURRING AT A SPECIFIC TIME, GIVEN THAT THE EVENT HAS NOT OCCURRED YET.
- ONCE THE HAZARD FUNCTION IS GENERATED, A **LOG-RANK STATISTICAL TEST** CAN BE USED TO HELP UNDERSTAND WHETHER THERE IS A STATISTICALLY SIGNIFICANT DIFFERENCE IN SURVIVAL.

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TIME-TO-EVENT ANALYSIS

- **KAPLAN-MEIER (KM) ESTIMATOR** – NON-PARAMETRIC MODEL THAT USES TIME-TO-EVENT DATA. TYPICALLY USED FOR CALCULATING SURVIVAL RATES OVER TIME FOR PATIENTS AFTER TREATMENT. IN THIS CONTEXT, RAINFALL IS CONSIDERED THE TREATMENT.
- **COX PROPORTIONAL HAZARDS MODEL** – COMPARABLE TO KM ESTIMATOR BUT ALLOWS FOR OTHER PREDICTOR VARIABLES TO BE CONSIDERED. VARIABLES COULD EITHER BE ASSOCIATED FASTER OR SLOWER TRAVEL TIMES.
- **GEOGRAPHICALLY STRATIFIED COX REGRESSION** – USES COX PROPORTIONAL HAZARDS MODEL AND **K-MEANS CLUSTERING**. K-MEANS CLUSTERING IS A MACHINE LEARNING ALGORITHM DESIGNED TO PARTITION DATA INTO K DIFFERENT CLUSTERS BASED ON SIMILARITY. THE PURPOSE IS TO ANALYZE HOW DIFFERENT LOCATIONS ARE AFFECTED.

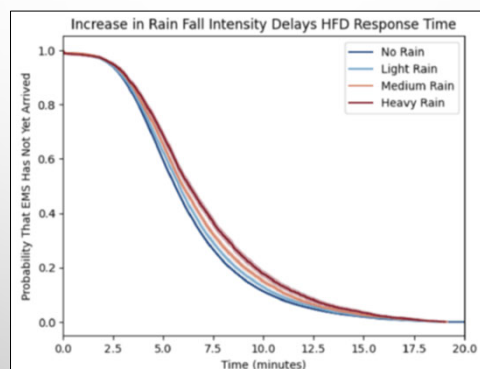
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REGRESSION

- **MULTIPLE LINEAR REGRESSION** – USED TO UNDERSTAND THE STRENGTH AND DIRECTION OF RELATIONSHIPS BETWEEN FEATURES AND TRAVEL TIME BUT ASSUMES A LINEAR RELATIONSHIP.
- **RANDOM FOREST** – ENSEMBLE LEARNING METHOD DESIGNED TO IMPROVE PREDICTIVE ACCURACY AND ROBUSTNESS BY COMBINING MULTIPLE DECISION TRAINING. EACH TREE IS TRAINED ON A RANDOM SUBSET OF THE TRAINING DATA WHILE FEATURES ARE ALSO RANDOMLY SELECTED DURING THE SPLITTING OF NODES IN A PROCESS REFERRED TO AS **BOOTSTRAP AGGREGATING**. USED FOR COMPLEX, NON-LINEAR RELATIONSHIPS. PROVIDES INSIGHTS ON WHICH PREDICTORS ARE THE MOST INFLUENTIAL. FOR THESE PURPOSES, RANDOM FOREST MODEL WAS TRAINED ON FEATURES SUCH AS UNIT TYPE, RAINFALL INTENSITY, INCIDENT TYPE, AND TIME OF DAY.

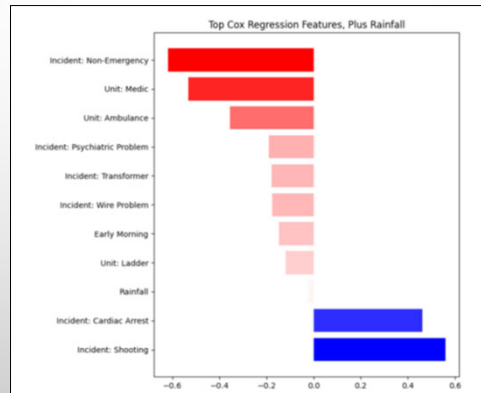
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FINDINGS: KM ESTIMATOR



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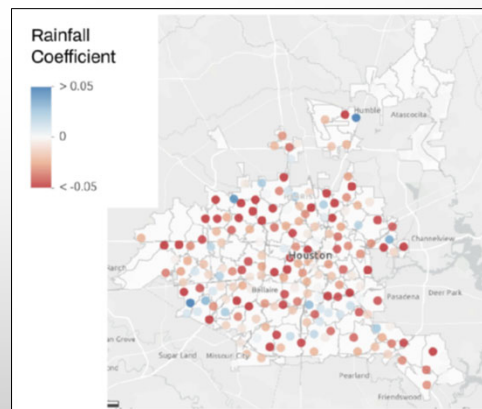
FINDINGS: COX PROPOSAL HAZARDS MODEL



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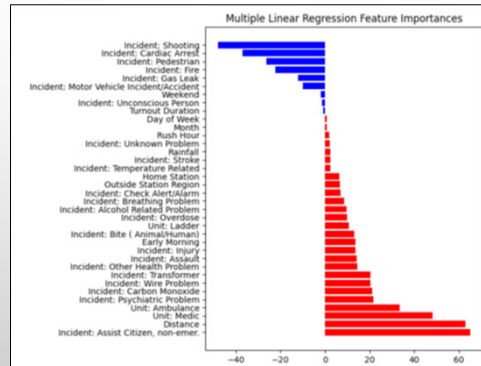
FINDINGS: COX PROPOSAL HAZARDS MODEL

- BLUE REPRESENTS FASTER RESPONSES.
- RED REPRESENTS DELAYED RESPONSES.



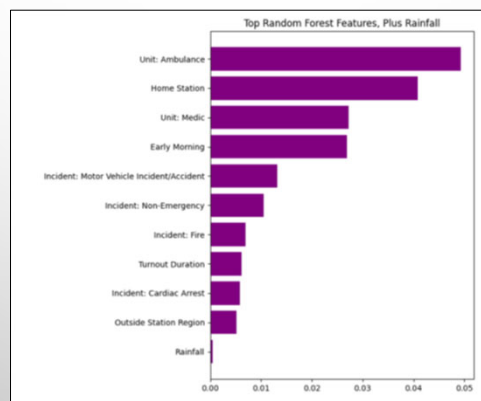
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FINDINGS: MULTIPLE LINEAR REGRESSION



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FINDINGS: RANDOM FOREST



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FALL 2024 D2K SHOWCASE



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Effects of Rainfall on Emergency Response Times for the Houston Fire Department

Jared Boyd¹, Cunyi Mao¹, Nico Motta¹, Lauren Yu¹, Leonard Chan², Xinjie Lan¹, Jiaming Liu¹
¹Rice University, ²Houston Fire Department



Background

- Houston Fire Department (HFD) responds to 300,000+ incidents each year
- Travel time- time between unit departing station and arriving at incident
 - Largest component of HFD response time
- Weather can lead to travel delays, leading to worse outcomes
 - Traffic, visibility, driving conditions

Objective

- Analyze the effect of rainfall and other variables on HFD travel time to understand causes of travel delays

Data

Incidents (HFD)

- Location
- Datetime
- Incident type

Unit Response (HFD)

- Unit type
- Responding station
- Travel time

Rainfall (Central Station, 1925a)

- Rain measured every 5 mins at 190 rainfall sites

Final Dataset

101,280 INCIDENTS
24 variables
Jan. 2022 - June 2024

Travel Time Difference in Rain vs. No Rain

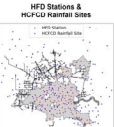
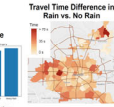
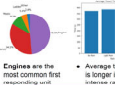
Unit Type

Engines are the most common first responding unit

Travel Time

Average travel time is longer in more rain events

Different stations are affected by rain to different extents

Methodology

Time-to-Event Analysis

Infer relationships with travel time



Kaplan-Meier Estimator

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right)$$

Where:

- $\hat{S}(t)$ is the probability of not reaching the emergency yet at t
- n_j is the number of emergencies at one has reached just before t_j
- d_j is the number of emergencies reached at t_j
- $t_j > 0, S(0) = 1$

Cox Proportional Hazards Model

$$h(t) = h_0(t) \cdot \exp(\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)$$

Where:

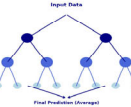
- $h_0(t)$ is the hazard rate at time t
- $h(t)$ is the baseline hazard function
- $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients associated with the covariates $X_1, X_2, \dots, X_k$
- $h_0(t)$ represents the exponential function

Regression

Predict travel time

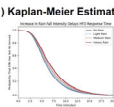
Random Forest

- Averages results from multiple decision trees to improve accuracy and reduce overfitting



Modeling Results

1) Kaplan-Meier Estimator



- As rain intensity increases, HFD is less likely to have arrived at a given point in travel time

2) Cox Proportional Hazards Model



- Confirms a statistically significant impact of rain delaying travel time
- Concordance: 0.73

3) Random Forest

- Tuned hyperparameters to minimize testing Root Mean Squared Error (RMSE)
- Number of trees: 500
- Maximum tree depth: 15

Performance

- R-Squared: 0.40
- RMSE: 1.4 minutes

Predictions within 1 minute	Predictions within 2 minutes
54%	88%

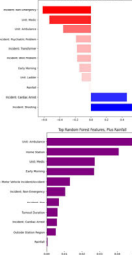
Feature Importances

Cox Regression

- Medic & Ambulance units take longer to arrive on scene, due to fewer availability, etc.
- Extreme incident types are the greatest accelerators of travel time
- Rainfall has a delay effect on travel time, but is less significant

Random Forest

- Confirms importance of unit type and incident type
- Station is an important predictor of travel time
- Early Morning impacts travel time, indicating operational timing is vital



Conclusion & Impact

- The models confirm the hypothesis that response times degrade in proportion to rainfall intensity
- Identifying the most impacted communities can help HFD provide faster support, especially with climate change causing more severe storms in Houston

Future Steps

- The low R^2 values leave room to identify more closely relevant variables that can improve model accuracy
- HFD can improve their data collection processes to make travel time estimations less susceptible to human error



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POTENTIAL

- FUTURE STUDIES THAT INJECT ADDITIONAL VARIABLES SUCH AS FLOOD PLAIN, ROADWAY, SCHOOL ZONE, ETC. DATA.
- IMPROVE ROBUSTNESS OF MODELS BY ESTABLISHING STARTING POINT OF UNITS AS ASSUMPTION MADE THAT ALL UNITS ARE STARTING FROM THEIR HOME STATION.
- ADJUST RESOURCE ALLOCATION TO ACCOUNT FOR RAIN-RELATED TRAVEL DELAYS.
- IMPROVE INFRASTRUCTURE WHERE TRAVEL TIMES ARE MOST AFFECTED BY RAINFALL.